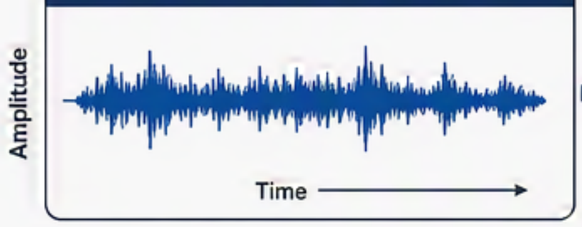
SOUND ANALYSIS PROCESSING CHAIN

1. RAW ACOUSTIC SIGNAL

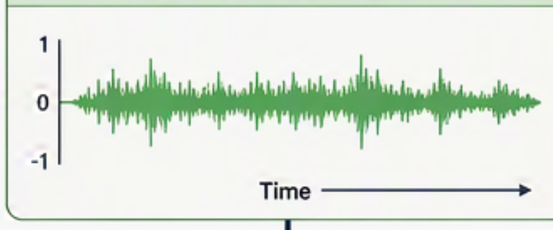


2. NORMALIZATION (Amplitude Scaling)

Scales signal to a consistent range



PRE-PROCESSED SIGNAL

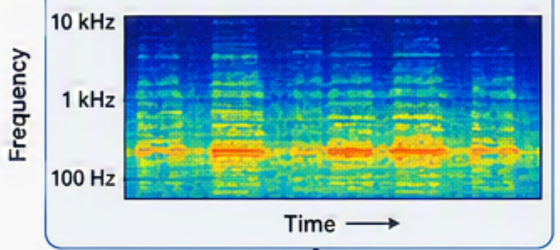


PRE-PROCESSING (typical)

- Filtering (band-pass, notch)
- Resampling
- Normalization

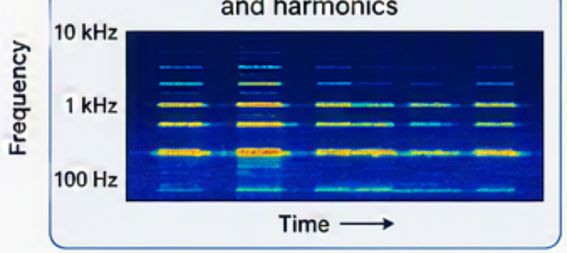
3. BROADBAND ANALYSIS (Energy View)

Wide frequency range
Shows overall energy distribution



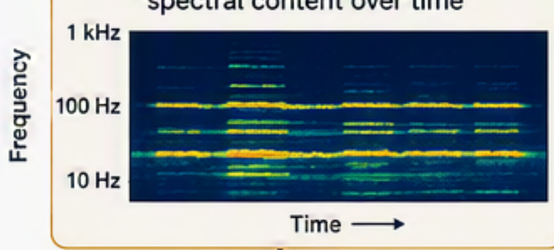
4. NARROWBAND ANALYSIS (Tonal Detail)

High frequency resolution
Reveals tonal components and harmonics



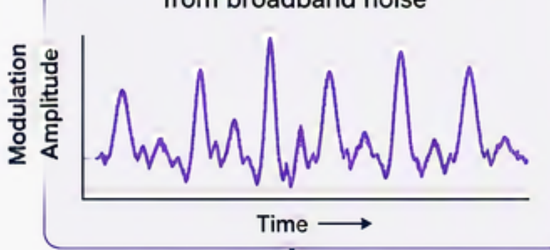
5. LOFAR ANALYSIS (Low Frequency Analysis and Recording)

Displays low frequency spectral content over time



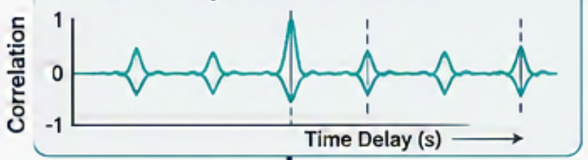
6. DEMON ANALYSIS (Detection of Envelope Modulation on Noise)

Extracts amplitude modulation from broadband noise



7. AUTO-CORRELATION (Periodicity Detection)

Measures similarity of signal with delayed versions of itself



8. FEATURE EXTRACTION (Rates and Frequencies)

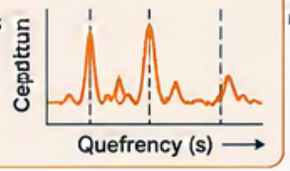
Estimates fundamental period, propeller rate, blade rate, etc.

- Shaft / propeller rate
- Blade passage frequency
- Modulation frequency



9. CEPSTRUM ANALYSIS (Hidden Periodic Structure)

Separates source characteristics from propagation effects and reveals hidden periodicities



10. SOURCE IDENTIFICATION & CLASSIFICATION

Combines extracted features to identify and classify the sound source

- Acoustic fingerprinting
- Target identification
- Classification / decision support



KEY

- Pre-processing
- Spectral Analysis
- Specialised Analysis
- Modulation Analysis
- Correlation / Feature
- Higher Level Analysis

WHAT EACH METHOD REVEALS

- **Broadband** – overall energy transients
- **Narrowband / LOFAR** – tonal frequencies and harmonics
- **DEMON** – amplitude modulation; rotating machinery
- **Auto-correlation** – periodicity, fundamental periods
- **Cepstrum** – hidden periodic structure; source characteristics

Together these techniques provide a complete picture of the acoustic signal, enabling detection, analysis and classification of sound sources.